

Complete Excel Notes: Basic to Advanced

Data Analyst Master Guide & Reference Handbook

PART 1 — EXCEL FUNDAMENTALS

1. What is Microsoft Excel?

Excel is spreadsheet software used to enter, organize, calculate, clean, analyze, and visualize data. A workbook can contain multiple worksheets, and each worksheet contains rows and columns made up of cells.

Typical Data Analyst Workflow:
Data Source → Data Preparation → Data Analysis → Data Visualization → Business Insight

2. Excel Structure

Term	Meaning	Example
Workbook	Complete Excel file	Sales.xlsx
Worksheet	Individual sheet inside workbook	Sales
Row	Horizontal series of cells	Row 5
Column	Vertical series of cells	Column C
Cell	Intersection of row and column	C5
Range	Group of cells	A2:D20
Active cell	Currently selected cell	B3
Formula Bar	View/edit cell contents	=SUM(B2:B10)
Name Box	Shows cell/range address	B3

3. Cell References

A cell reference identifies a cell. Reference types determine how formulas behave when copied to other cells.

Reference	Example	Behavior when copied
Relative	A1	Both row and column update
Absolute	\$A\$1	Both row and column stay fixed
Mixed Column	\$A1	Column fixed; row updates
Mixed Row	A\$1	Row fixed; column updates
Range	A1:C10	Refers to a block of multiple cells

Practical Example (Tax Formula): If B2 contains sales and E1 contains a tax rate, use =B2*\$E\$1. Copying the formula downward changes B2 to B3, B4, etc., while \$E\$1 remains completely fixed.

4. Data Types

- **Text:** Product, Pune, Employee Name

- **Number:** 100, 2500.50
- **Date:** 18/08/2026
- **Time:** 10:30 AM
- **Percentage:** 15%
- **Currency:** \$50,000 / ₹50,000
- **Logical:** TRUE or FALSE
- **Formula:** Starts with = and calculates a dynamic result

5. Basic Formatting Options

Feature	Primary Purpose
Bold / Font	Emphasize headings and key summary values
Fill Color	Highlight distinct table sections or key KPI cards
Borders	Visually separate table regions, headers, and totals
Alignment	Improve visual scanning and alignment across metrics
Wrap Text	Display multiline text cleanly inside a single cell
Merge & Center	Useful for main report titles; avoid using on raw data tables
Number Format	Standardize display of currency, percentages, and dates
Freeze Panes	Keep top headers or key columns visible while scrolling

PART 2 — FORMULAS & FUNCTIONS

Formula Basics

A formula begins with =. Operators include + (addition), - (subtraction), * (multiplication), / (division), ^ (exponent), and & (text joining).

- **Add:** =A2+B2
- **Subtract:** =A2-B2
- **Multiply:** =A2*B2
- **Divide:** =A2/B2
- **Percentage Ratio:** =B2/A2
- **Growth %:** =(B2-A2) / A2
- **Text Join:** =A2&" "&B2

6. Core Math & Statistical Functions

Function	Syntax Example	Primary Purpose
SUM	=SUM(B2:B10)	Total of numeric range
AVERAGE	=AVERAGE(B2:B10)	Arithmetic mean
MIN	=MIN(B2:B10)	Smallest value in range

Function	Syntax Example	Primary Purpose
MAX	=MAX(B2:B10)	Largest value in range
COUNT	=COUNT(B2:B10)	Counts numeric cells only
COUNTA	=COUNTA(A2:A10)	Counts all non-empty cells
COUNTBLANK	=COUNTBLANK(A2:A10)	Counts empty/blank cells
MEDIAN	=MEDIAN(B2:B10)	Middle numeric value
MODE.SNGL	=MODE.SNGL(B2:B10)	Most frequently occurring value
STDEV.S	=STDEV.S(B2:B10)	Sample standard deviation
RANK.EQ	=RANK.EQ(B2,\$B\$2:\$B\$10,0)	Rank of value within a list

7. Conditional Functions

Function	Formula Example	Operational Logic
IF	=IF(C2>=50,"Pass","Fail")	Evaluates a single logical condition
IFS	=IFS(C2>=80,"A",C2>=60,"B",TRUE,"C")	Evaluates multiple condition branches
AND	=AND(B2>=50,C2>=50)	Returns TRUE if all arguments are true
OR	=OR(B2>=50,C2>=50)	Returns TRUE if any argument is true
NOT	=NOT(A2="Yes")	Reverses logical value (TRUE → FALSE)
IFERROR	=IFERROR(A2/B2,0)	Catches errors and returns custom default

8. Criteria-Based Aggregations

Function	Syntax Example	Usage Scenario
SUMIF	=SUMIF(A2:A100,"Pune",B2:B100)	Sum values based on single condition
SUMIFS	=SUMIFS(D2:D100,A2:A100,"Pune",B2:B100,"Laptop")	Sum values based on multiple conditions
COUNTIF	=COUNTIF(A2:A100,"Pune")	Count records matching single condition
COUNTIFS	=COUNTIFS(A2:A100,"Pune",C2:C100,"Male")	Count records matching multiple conditions
AVERAGEIF	=AVERAGEIF(A2:A100,"Pune",D2:D100)	Average values matching single condition
AVERAGEIFS	=AVERAGEIFS(D2:D100,A2:A100,"Pune",B2:B100,"Laptop")	Average values matching multiple conditions

PART 3 — LOOKUPS & REFERENCE FUNCTIONS

9. VLOOKUP

Searches the leftmost column of a table and retrieves a matching value from a specified column to the right.

- **Syntax:** =VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
- **Example:** =VLOOKUP(E2, A2:C100, 3, FALSE)
- **Limitations:** Can only search left-to-right from the initial lookup column and relies on hardcoded column index numbers.

10. XLOOKUP

Modern, flexible lookup function that replaces VLOOKUP and HLOOKUP.

- **Syntax:** =XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found])
- **Example:** =XLOOKUP(E2, A2:A100, C2:C100, "Not Found")
- **Key Advantages:** Looks left or right, handles missing records natively, and defaults to exact match.

11. INDEX + MATCH / XMATCH

- **INDEX Function:** Returns a value located at a specific row/column position.
- **MATCH Function:** Returns the numeric position of an item in a range.
- **Combined Formula:** =INDEX(C2:C100, MATCH(E2, A2:A100, 0))
- **Modern Alternative (XMATCH):** =XMATCH(E2, A2:A100, 0)

12. Text Functions

Function	Example	Description / Purpose
LEFT	=LEFT(A2,5)	Extract specified number of starting characters
RIGHT	=RIGHT(A2,4)	Extract specified number of ending characters
MID	=MID(A2,3,5)	Extract characters starting from middle position
LEN	=LEN(A2)	Count total character count of a string
TRIM	=TRIM(A2)	Strip extra spaces except single spaces between words
CLEAN	=CLEAN(A2)	Strip non-printable characters
UPPER	=UPPER(A2)	Convert text to all uppercase
LOWER	=LOWER(A2)	Convert text to all lowercase
PROPER	=PROPER(A2)	Convert text to title case
CONCAT	=CONCAT(A2,B2)	Join string values without delimiters
TEXTJOIN	=TEXTJOIN("-",TRUE,A2:C2)	Join cells with specified delimiter, skipping blanks
SUBSTITUTE	=SUBSTITUTE(A2,"old","new")	Replace occurrences of specific text string
FIND	=FIND("@",A2)	Case-sensitive character search
SEARCH	=SEARCH("@",A2)	Case-insensitive character search

13. Date & Time Functions

Function	Example Formula	Primary Purpose
TODAY	=TODAY()	Returns current date
NOW	=NOW()	Returns current date and timestamp
DATE	=DATE(2026,8,18)	Construct valid date from year, month, day
YEAR / MONTH / DAY	=YEAR(A2)	Extract date components into integers
EOMONTH	=EOMONTH(A2,0)	Returns last day of the month
WEEKDAY	=WEEKDAY(A2,2)	Day-of-week integer index
WORKDAY	=WORKDAY(A2,10)	Date after specified working days
NETWORKDAYS	=NETWORKDAYS(A2,B2)	Total working days between two dates

PART 4 — ADVANCED EXCEL

14. Dynamic Array Functions

Dynamic array formulas return multiple values across adjacent cells automatically (spilling).

Function	Formula Example	Operational Purpose
FILTER	=FILTER(A2:D100,D2:D100="Pune")	Extract dynamic subset matching criteria
SORT	=SORT(A2:D100,2,-1)	Sort array by column index and order
SORTBY	=SORTBY(A2:D100,D2:D100,-1)	Sort array using a separate reference range
UNIQUE	=UNIQUE(A2:A100)	Extract distinct list of values
SEQUENCE	=SEQUENCE(10,1,1,1)	Generate sequential list of numbers
TRANSPOSE	=TRANSPOSE(A2:D2)	Flip orientation from row to column or vice versa

15. Advanced & Optimization Functions

- **LET Function:** =LET(x, B2*C2, x*10%)
Assigns variable names to intermediate calculations to boost calculation performance and readability.
- **LAMBDA Function:** =LAMBDA(x, x*10%)
Creates custom reusable functions without writing VBA code.
- **SUMPRODUCT Function:** =SUMPRODUCT(B2:B10, C2:C10)
Multiplies corresponding ranges and sums the calculated products.
- **SUBTOTAL Function:** =SUBTOTAL(9, B2:B100)
Calculates aggregates while ignoring rows hidden by auto-filters.
- **AGGREGATE Function:** =AGGREGATE(9, 6, B2:B100)
Flexible aggregate calculations with options to ignore hidden rows and error values.

16. Tables & Structured References

Select dataset and press `Ctrl + T`. Tables expand dynamically, retain formatting, integrate auto-filters, and support structured reference syntax.

Example: `=SUM(Sales[Amount])` sums the Amount column from the structured table named Sales.

17. Conditional Formatting

Highlight key insights automatically, such as duplicates, top performers, overdue dates, or entire conditional rows.

Formula-Based Rule Example: `=$E2="Overdue"`

PART 5 — DATA ANALYSIS & VISUALIZATION

18. PivotTables

PivotTables allow interactive aggregation of complex datasets. Fields can be placed into four distinct drag-and-drop areas: Rows, Columns, Values, and Filters. Aggregate metrics include Sum, Count, Average, Min, and Max.

Layout Example:
Rows: Region | Columns: Month | Values: Sum of Sales | Filters: Product

19. Slicers & PivotCharts

- **Slicers:** Clickable UI visual buttons that filter attached PivotTables and PivotCharts.
- **PivotCharts:** Dynamic chart elements linked directly to PivotTable outputs.

20. Chart Selection Matrix

Business Query	Recommended Visualization
Compare categories	Bar / Column Chart
Show trend over time	Line Chart
Show relationship between two numeric variables	Scatter Plot
Show contribution to a whole	Pie / Doughnut Chart (few categories only)
Show positive/negative components of a total	Waterfall Chart
Show frequency distribution	Histogram

21. KPI & Dashboard Design

- **Key Performance Indicators (KPIs):** High-level operational metrics like Revenue, Profit Margin %, Order Volume, and Conversion Rate.
- **Dashboard Principles:** Combine consolidated KPI cards, concise visualizations, interactive slicer filters, and underlying summaries to communicate performance clearly.

PART 6 — DATA CLEANING & POWER QUERY

22. Data Cleaning Best Practices

- Remove duplicate entries systematically.
- Treat missing data intentionally—avoid filling blanks with zeros indiscriminately.

- Standardize text values via TRIM, CLEAN, and casing functions (UPPER, PROPER).
- Enforce strict data type definitions across columns.
- Validate date ranges, product keys, and key values.
- Isolate and audit extreme outliers.
- Preserve pristine raw source files separate from modified pipelines.

23. Power Query Pipeline (ETL)

Power Query enables automated, repeatable Data Import, Transformation, and Loading (ETL).

Import Data → Change Types → Filter Rows → Split/Merge → Group & Pivot → Load & Refresh

- **Merge:** Joins tables side-by-side using key identifiers (similar to database SQL joins).
- **Append:** Stacks tables vertically with identical column header structures.

PART 7 — POWER PIVOT, DATA MODEL & DAX

24. Power Pivot & Data Model

Power Pivot links multiple tables via direct key relationships using a Star Schema layout.

- **Fact Table:** Central transactions table (e.g., Sales Transactions).
- **Dimension Tables:** Surrounding lookup context tables (e.g., Date, Product, Customer, Region).

25. Core DAX Formulas

Metric Name	DAX Expression Syntax	Description
Total Sales	Total Sales = SUM(Sales[Amount])	Aggregates gross sales value
Total Orders	Total Orders = COUNTROWS(Sales)	Counts transaction records
Unique Customers	Customers = DISTINCTCOUNT(Sales[CustomerID])	Counts distinct customer keys
Profit	Profit = SUM(Sales[Sales]) - SUM(Sales[Cost])	Calculates total net profit
Profit Margin	Profit Margin = DIVIDE([Profit], [Total Sales], 0)	Evaluates ratio safely

Key DAX Modifiers:

- CALCULATE (. . .) : Modifies filter context for calculations.
- FILTER (. . .) : Returns a filtered table expression.
- ALL (. . .) : Ignores active filters.
- RELATED (. . .) : Fetches values across active relationships from dimension tables.

26. What-If Analysis & Forecasting

Analysis Tool	Operational Application
Goal Seek	Adjusts single input value to hit specific calculated target output
Scenario Manager	Defines, switches, and compares multi-variable assumptions
Data Table	Evaluates 1-variable or 2-variable sensitivity grids

Analysis Tool	Operational Application
Forecast Sheet	Generates statistical time-series projections based on historical data
Trendline	Overlays linear/exponential regression equations on visual charts

PART 8 — REAL DATA ANALYST EXAMPLES

Example 1: Sales Performance Dashboard

- Metrics:**

$$Revenue = SUM(Sales[Sales])$$

$$Profit = SUM(Sales[Sales]) - SUM(Sales[Cost])$$

$$Profit\ Margin = Profit / Revenue$$

$$Average\ Order\ Value = Revenue / Total\ Orders$$
- Workflow:** Group metrics by Region, Product, and Month using PivotTables, link PivotCharts, and build an executive summary view.

Example 2: Employee HR Records Analysis

Workflow: Evaluate service tenure, flag missing employee records, categorize operational departments, compute median salaries, and rank organizational performance.

Example 3: Customer Segmentation

Workflow: Apply COUNTIFS and SUMIFS to group customer brackets, enrich profile details via XLOOKUP, and aggregate spend categories using PivotTables.

Example 4: Automated Text Cleaning

- Problem:** Text field contains untrimmed text with inconsistent casing: ' pune '
- Solution:** =PROPER(TRIM(A2)) returns ' Pune ' clean and formatted. Automate recurring file imports via Power Query.

27. Key Shortcuts Cheat Sheet

Keyboard Shortcut	Executed Action
Ctrl + C / V / X	Copy / Paste / Cut selected cells
Ctrl + Z / Y	Undo / Redo action
Ctrl + S	Save active workbook
Ctrl + F / H	Find / Replace dialog
Ctrl + T	Convert selected range to Excel Table
Ctrl + Shift + L	Toggle filters on/off
Alt + =	Insert AutoSum formula
F2	Edit active cell
Ctrl + 1	Open Format Cells dialog
Ctrl + Arrow Key	Jump to edge of data region

Keyboard Shortcut	Executed Action
Ctrl + Shift + Arrow	Select region to edge of data
Ctrl + PageUp / PageDown	Navigate between worksheets

28. Troubleshooting Common Errors

Error Code	Cause & Fix Strategy
#DIV/0!	Attempted division by zero; validate inputs or wrap with IFERROR
#N/A	Lookup value not found; verify lookup keys, extra spaces, or match modes
#VALUE!	Incompatible data types used in mathematical operation
#REF!	Invalid or deleted cell reference
#NAME?	Misspelled function name or unquoted text inside formula
#NUM!	Invalid numeric calculation or parameter value out of bounds
#####	Column width is too narrow to display value, or output is a negative date

PART 9 — JOB-READY EXCEL ROADMAP

- [Level 1: Workbook Core & Data Types]
- ★
- [Level 2: Tables, Formatting & Filters]
- ★
- [Level 3: Conditional Logic & Basic Aggregations]
- ★
- [Level 4: Advanced Lookups (XLOOKUP) & Date/Text Cleaning]
- ★
- [Level 5: PivotTables, Slicers & Dynamic Formatting]
- ★
- [Level 6: Analytical Reporting Pipelines]
- ★
- [Level 7: Dynamic Array Functions (FILTER, UNIQUE, SORT)]
- ★
- [Level 8: Power Query & Automated ETL]
- ★
- [Level 9: Power Pivot Data Modeling & DAX Measures]
- ★
- [Level 10: Executive Dashboards & Strategic Storytelling]

Interview Questions to Practice

- What is the difference between a workbook and a worksheet?
- Explain relative vs absolute cell references with an example.
- How does VLOOKUP compare to XLOOKUP?
- Compare INDEX-MATCH versus XLOOKUP.
- When do you use SUMIF vs SUMIFS?
- How does COUNT differ from COUNTA?

- What is your workflow to remove duplicate records and handle missing values?
- In what scenarios would you build a PivotTable?
- What is Power Query and why is it valuable for automated analysis?
- Explain Merge vs Append operations in Power Query.
- What is a Data Model in Excel?
- Define DAX and explain what a calculated measure is.
- How would you design an end-to-end sales performance dashboard?
- How do you validate and audit your analytical models for accuracy?